

Troubleshooting Guide: Ammonia Refrigeration System — Oil Fouling Reducing Evaporator Capacity

Category: Utilities / Industrial Refrigeration **Unit:** Ammonia Refrigeration Package, Flooded Evaporator **Tools:** Refrigerant-side heat transfer performance trending, compressor oil separator efficiency check, oil sample analysis **Fluid Package:** Not applicable in the VLE sense — ammonia refrigeration cycle performance uses standard refrigerant property tables, not a hydrocarbon cubic EOS **Symptom:** Process outlet temperature gradually rising, compressor amperage climbing, suction pressure trending down slightly, at unchanged load

1. Quick Reference

Item	Detail
Reported symptom	Process outlet temperature creeping up; compressor amps rising; suction pressure trending down slightly — all at unchanged process load
Initially unclear	Whether this was a refrigerant charge/leak issue, a compressor mechanical issue, or a heat transfer degradation inside the evaporator
Actual root cause	Compressor oil separator demister pad had degraded, allowing increased oil carryover into the evaporator, where it formed an insulating film on the tube surface and reduced heat transfer
Fix	Replaced oil separator demister pad; drained accumulated oil from the evaporator low point
Diagnostic signal	Refrigerant-side U calculated from actual data had dropped well below clean design value, while refrigerant charge and compressor mechanical checks came back normal
Prevention	Periodic oil level trending at the evaporator low point; scheduled oil separator efficiency testing

2. Symptom

- **Process outlet temperature gradually rose** at unchanged process load — the evaporator wasn't removing as much heat as before.
- **Compressor amperage climbed** and **suction pressure trended down slightly**, both consistent with the compressor working harder to maintain the same duty.

3. Why This Wasn't Assumed to Be a Simple Charge or Compressor Issue

A slowly declining refrigeration performance is often first attributed to a **slow refrigerant leak** (reducing charge) or a **compressor mechanical issue** (reduced volumetric efficiency, similar in principle to the reciprocating compressor case). Both are common and both were worth ruling out before considering a heat-transfer-side cause, since a heat-transfer degradation points toward a completely different fix — cleaning the evaporator, not chasing a leak or overhauling the compressor.

4. Diagnostic Approach

Step 1 — Confirm refrigerant charge

Refrigerant charge level was checked via standard system indicators (receiver level, sight glass) and found within normal range — ruling out a leak as the primary driver.

Step 2 — Confirm compressor mechanical performance

Compressor discharge pressure, suction pressure, and amperage were reviewed against the compressor's performance curve for current conditions; the compressor itself was found to be operating consistently with its curve — not showing a mechanical efficiency loss of its own.

Step 3 — Calculate actual evaporator heat transfer performance

With charge and compressor both ruled out, the evaporator's **actual overall heat transfer coefficient** was calculated from measured refrigerant and process-side temperatures and flows, and compared against the clean/design value — the same core technique used for the seawater cooler and crude preheat train cases.

Finding: Actual U had dropped **well below clean design**, confirming a heat-transfer-side degradation specifically at the evaporator.

Step 4 — Investigate the fouling mechanism

With reduced U confirmed, the compressor's **oil separator** was inspected, since oil carry-over into a flooded evaporator is a well-known cause of reduced heat transfer (oil forms an insulating film on the tube/shell surface, since oil has much lower thermal conductivity than ammonia liquid).

Finding: The oil separator's **demister pad had degraded**, reducing its oil removal efficiency and allowing more oil to carry over into the evaporator.

Quantitative Basis

- Evaporator clean design U: 165 Btu/hr·ft²·°F. Trended actual U dropped to **98 Btu/hr·ft²·°F — a 41% loss.**
- Process outlet temperature rose from a design 20°F to **28°F** at unchanged load; compressor amperage rose from a baseline 145 A to **172 A.**
- Oil separator efficiency (rated to limit carryover to <5 ppm at the compressor discharge) was measured at the degraded demister pad passing **340 ppm** — a 68x exceedance.
- Oil drained from the evaporator low point measured **11 gallons**, versus a normal trace/negligible accumulation for a properly functioning oil separator.

5. Root Cause

The compressor's oil separator demister pad had degraded, significantly increasing oil carryover into the evaporator. The carried-over oil accumulated on the evaporator's heat transfer surface, forming an insulating film that reduced the effective overall heat transfer coefficient — reducing cooling capacity at unchanged process load and forcing the compressor to work harder (higher amps, lower suction pressure) to try to compensate.

6. Corrective Action

1. **Replaced the oil separator demister pad**, restoring its oil removal efficiency.
2. **Drained the accumulated oil from the evaporator low point.**

7. Verification

- Oil separator efficiency, re-tested post-repair, measured **3 ppm carryover**, within the <5 ppm design target.
- Evaporator U recovered to **158 Btu/hr·ft²·°F — 96% of clean design.**
- Process outlet temperature returned to **21°F**, and compressor amperage dropped to **148 A**, both close to baseline, and held stable over the following **30 days.**

8. Prevention / Long-Term Fix

- Established **periodic oil level trending at the evaporator low point**, so future oil accumulation is caught early rather than discovered only through a broader performance decline.
- Implemented **scheduled oil separator efficiency testing**, since a demister pad can degrade gradually without triggering any single obvious alarm.

9. General Troubleshooting Checklist (Reusable)

- ☐ Confirm refrigerant charge level before assuming a heat-transfer-side cause
- ☐ Confirm compressor performance against its rated curve for current suction/discharge conditions
- ☐ If both check out normal, calculate actual evaporator U from measured data and compare to clean design U
- ☐ If U has dropped, inspect the compressor's oil separator (demister condition, measured carryover ppm)
- ☐ Drain and measure any accumulated oil at the evaporator low point as direct physical confirmation
- ☐ Replace/repair the oil separator and drain accumulated oil
- ☐ Confirm recovery via BOTH oil separator carryover test AND evaporator U, not just a general temperature improvement
- ☐ Establish routine oil level trending at evaporator low points and scheduled oil separator efficiency testing

10. Key Takeaway

In flooded ammonia (or other oil-lubricated refrigerant) systems, a gradually declining evaporator performance with normal charge and normal compressor mechanicals should point straight at oil carryover — oil is an effective insulator on a heat transfer surface, and a degrading oil separator can quietly rob capacity long before

any single alarm fires. Calculate actual U against clean design, and if it's degraded, check the oil separator before assuming a leak or a compressor problem.

Related Concepts / Tags

ammonia-refrigeration flooded-evaporator oil-carryover oil-separator demister-pad
heat-transfer-coefficient industrial-refrigeration U-value-trending

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.